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This work done is stored as the electrostatic potential energy U of a system of charges $q_1$ and $q_2$ separated by a distance r. Thus we have

$$U = q_2 V = \frac{1}{4\pi\varepsilon_\circ}\frac{q_1 q_2}{r} \qquad (1.45)$$

The electrostatic potential energy depends only on the distance between the two point charges. In fact, the expression (1.45) is derived by assuming that $q_1$ is fixed and $q_2$ is brought from infinity. The equation (1.45) holds true when $q_2$ is fixed and $q_1$ is brought from infinity or both $q_1$ and $q_2$ are simultaneously brought from infinity to a distance *r* between them.

Three charges are arranged in the following configuration as shown in Figure 1.30.

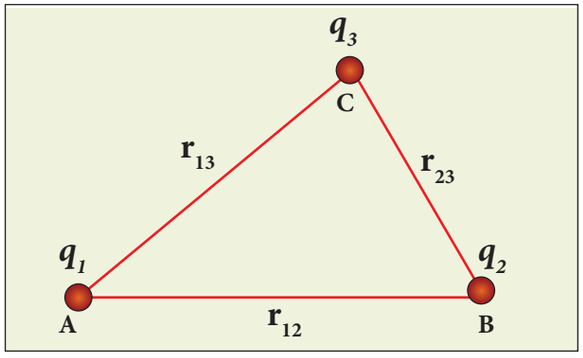



**Figure 1.30** Electrostatic potential energy for Collection of point charges

To calculate the total electrostatic potential energy, we use the following procedure. We bring all the charges one by one and arrange them according to the configuration as shown in Figure 1.30.

**(i)** Bringing a charge $q_1$ from infinity to the point A requires no work, because there are no other charges already present in the vicinity of charge $q_1$.

**(ii)** To bring the second charge $q_2$ to the point B, work must be done against the electric field created by the charge $q_1$. So the work done on the charge $q_2$ is $W = q_2 V_{1B.}$ Here $V_{1B}$ is the electrostatic potential due to the charge $q_1$ at point *B*.

$$U = \frac{1}{4\pi\varepsilon_\circ}\frac{q_1 q_2}{r_{12}} \qquad (1.46)$$

Note that the expression is same when $q_2$ is brought first and then $q_1$ later.

**(iii)** Similarly to bring the charge $q_3$ to the point C, work has to be done against the total electric field due to both charges $q_1$ and $q_2$. So the work done to bring the charge $q_3$ is = $q_3$ ($V_{1C}$ + $V_{2C}$). Here $V_{1C}$ is the electrostatic potential due to charge $q_1$ at point C and $V_{2C}$ is the electrostatic potential due to charge $q_2$ at point C.

The electrostatic potential is

$$U = \frac{1}{4\pi\varepsilon_\circ}\left(\frac{q_1 q_3}{r_{13}} + \frac{q_2 q_3}{r_{23}}\right) \qquad (1.47)$$

**(iv)** Adding equations (1.46) and (1.47), the total electrostatic potential energy for the system of three charges $q_1$, $q_2$ and $q_3$ is

$$U = \frac{1}{4\pi\varepsilon_\circ}\left(\frac{q_1 q_2}{r_{12}} + \frac{q_1 q_3}{r_{13}} + \frac{q_2 q_3}{r_{23}}\right) \qquad (1.48)$$

Note that this stored potential energy U is equal to the total external work done to assemble the three charges at the given locations. The expression (1.48) is same if the charges are brought to their positions in any other order. Since the Coulomb force is a conservative force, the electrostatic potential energy is independent of the manner in which the configuration of charges is arrived at.



UNIT-1(XII-Physics_Vol-1).indd 35 04-03-2019 10:25:51